

Sophie Ruehr
 Biosphere Sciences & Engineering Division
 Carnegie Institution for Science
 Stanford, CA USA

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ACADEMIC APPOINTMENTS

Carnegie Institution for Science

2025-	Postdoctoral Fellow in Land-Surface Modeling Biosphere Sciences & Engineering Division	Stanford, CA
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EDUCATION

Ph.D.	University of California Berkeley	Environmental Science, Policy & Management	2025
B.Sc.	Yale University	Geology & Geophysics	2018

RESEARCH INTERESTS

AI for Earth data, ecohydrology, proximal & satellite hyperspectral remote sensing, sensor development, climate solutions, drought resilience, sustainable agriculture, terrestrial ecosystems, water management, land-atmosphere interactions, stakeholder perspectives & policy implications

PROFESSIONAL EXPERIENCE

Provincetown Independent

2019-2020	Newspaper Staff Reporter	Provincetown, MA USA
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University of the South Pacific

2018-2019	Yale University Huang Fellow	Port Vila, Vanuatu
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Woods Hole Oceanographic Institution

2017	Summer Student Fellow	Woods Hole, MA USA
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Provincetown Banner

2016	Newspaper Staff Reporter	Provincetown, MA USA
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HONORS & AWARDS

Max Plank-Caltech-Carnegie-Columbia MC³ 4 Earth Center Post-Doctoral Fellowship, 2025-present
 NASA Future Investigators in Earth and Space Science and Technology Fellowship, 2022-2025
 Achievement Rewards for College Scientists Fellowship, 2020-2022
 National Science Foundation Graduate Research Fellowship Program Honorable Mention, 2022
 New England Newspaper Association, First Place: Science/Technology Reporting, 2021
 New England Newspaper Association, First Place: Health Reporting, 2021
 Yale University Parker Huang Undergraduate Travel Fellowship, 2018-2019
 Yale University Dept. of Geology & Geophysics Hammer Prize, 2018
 Yale University Karen Von Damm 1977 Fellowship, 2017
 Woods Hole Oceanographic Institution Summer Student Fellowship, 2017

GRANTS & COMPETITIVE FUNDING AWARDS (\$450,000 total)

2025-	Max Plank-Caltech-Carnegie-Columbia MC ³ 4 Earth Center: Postdoctoral Fellowship in Land-Surface Modeling (\$82,500 annually, up to 4 years)
2025-2026	University of California Berkeley Chancellor's Advisory Committee: Improving field safety and inclusion in ESPM (\$54,000)
2025-2026	University of California Berkeley Be Smart About Safety Grant: Field safety through improved emergency communications and training (\$25,000)
2022-2025	NASA FINESST (Future Investigators in NASA Earth and Space Science and Technology): Quantifying ecosystem reliance on groundwater, Co-PI Ruehr (\$150,000)
2022	FLUXNET Early Career Secondment: Quantifying ecosystem reliance on groundwater with eddy covariance and sap flow data (\$7,400)
2020	University of California Berkeley Carol Baird Fieldwork Grant: Using hyperspectral imagery at Sagehen Experimental Forest to quantify ecosystem groundwater use (\$33,000)
2020	Achievement Rewards for College Scientists Graduate Student Fellowship: Graduate funding for 2 years (\$100,000)
2018	Yale University Parker Huang Undergraduate Travel Fellowship: Oral history in Vanuatu to support paleoclimate research (\$36,000)
2017	Yale University Karen Von Damm 1977 Fellowship: In support of senior honors thesis field research in Lanzhou, China (\$5,000)
2017	Woods Hole Oceanographic Institution Summer Student Fellowship: Hurricane paleoclimatology research (\$8,000)

SERVICE***Committees & management***

UC Berkeley Dept. ESPM Field Safety Committee, 2024-2025

UC Berkeley College of Natural Resources LGBTQ+ Coalition, 2020-2023

AmeriFlux Diversity, Equity & Inclusion Committee, 2020-2024

UC Berkeley Dept. ESPM Graduate Diversity Council, 2020-2024

Reviewing

Regular reviewer for Nature Communications, PNAS, Science Advances, One Earth, Geophysical Research Letters, Agricultural & Forest Meteorology, Nature Communications Earth & Environment, Hydrology, Earth's Future, AGU Advances, & Journal of Arid Environments

MENTORING***Current***

Eden Gonzalez, UC Berkeley undergraduate student, 2023-

Jackson Coldiron, UC Santa Barbara master's student, 2024-

Past

Adam Rashid, UC Berkeley undergraduate student (now: MIT PhD student), 2023-2024

Megan Hur, UC Berkeley undergraduate student (now: NASA Goddard research assistant), 2022-2024

Tyler Goldstein, UC Berkeley undergraduate (now: UC Berkeley science communications), 2021-2023

TEACHING***Guest lectures***

- 2025 “Land Surface Modeling” in ESPM 111, Carbon Cycle Dynamics, UC Berkeley
 2024 “Remote Sensing of the Biosphere” in EPS 251, Ecosystem Science, UC Berkeley

Courses

- 2024 Graduate Student Instructor, Ecosystem Science, led by Prof. Dennis Baldocchi

Centers

- 2020-2024 Data Consultant for R, Python, Google Earth Engine, UC Berkeley D-Lab

PUBLICATIONS (see [Google Scholar Profile](#) for updates)***Published***

- **Ruehr S**, Pierrat Z, Parazoo N, Keenan TF. 2026. Harnessing solar-induced fluorescence for agricultural research and management. *Environmental Research Letters*. 10.1088/1748-9326/ae74e2.
- Rao et al. [29 authors incl. **Ruehr S**]. 2026. Decoupled carbon assimilation and growth responses to aridity in temperate deciduous oaks. *Science Advances*. 10.1126/sciadv.ady7139.
- Ospina D et al. [70 authors incl. **Ruehr S**]. 2026. Ten new insights in climate science 2026. *Global Sustainability*, 9:e6. 10.1017/sus.2025.10043.
- Pierrat ZA, Gustine RN, Boser A, **Ruehr S**, Lee CM, Reager JT, Cawse-Bicholson K. 2026. Human contributions to evapotranspiration mitigate swings in dry-to-wet year transitions. *Communications Sustainability*, 1, 8. 10.1038/s44458-025-00002-w.
- **Ruehr S**, Bassiouni M, Kang Y, Socolar Y, Magney T, Keenan TF. 2025. Crop diversification improves water-use efficiency and regional water sustainability. *Environmental Research Letters*, 20, 114062. 10.1088/1748-9326/ae15a9.
- **Ruehr S**, Gerlein-Safdi C, Falco N, Seibert P, Chou C, Albert L, Keenan TF. 2024. Quantifying seasonal and diurnal cycles of solar-induced fluorescence. *Geophysical Research Letters*, 51, 14. 10.1029/2023GL107429.
- **Ruehr S**, Keenan TF, Williams C, Zhou Y, Lu X, Bastos A, Canadell P, Prentice IC, Sitch S, Terrer C. 2023. Evidence and attribution of the enhanced land carbon sink. *Nature Reviews Earth & Environment*, 4, 518-534. 10.1038/s43017-023-00456-3.
- **Ruehr S**, Giroto G, Verfaillie J, Baldocchi D, Cabon A, Keenan TF. 2023. Ecosystem groundwater use enhances carbon sinks in a semi-arid oak savanna. *Agricultural & Forest Meteorology*, 342, 109725. 10.1016/j.agrformet.2023.109725.
- Massoud EC, Andrews L, Reichle R, Molod A, Park J, **Ruehr S**, Giroto M. 2022. Seasonal forecasting skill for the High Mountain Asia region in the Goddard Earth Observing System. *Earth System Dynamics*, 14, 147–171. 10.5194/esd-14-147-2023.
- **Ruehr S**. 2021. Beyond the vulnerability/resilience dichotomy: Perceptions of and responses to the climate crisis on Emau, Vanuatu. *Island Studies Journal*, 10.24043/isj.151.

- **Ruehr S**, Lee X, Smith R, Li X, Xu Z, Liu S, Yang X, Zhou Y. 2020. A mechanistic investigation of the oasis effect in the Zhangye cropland in semiarid western China. *Journal of Arid Environments*, 176, 104120. 10.1016/j.jaridenv.2020.104120.
- Espeland M et al. [14 authors incl. **Ruehr S**]. 2015. Ancient Neotropical origin and recent recolonization: Phylogeny, biogeography and diversification of the Riodinidae (Lepidoptera: Papilionoidea). *Molecular Phylogenetic Evolution*, 93, 296-306. 10.1016/j.ympev.2015.08.006.

In review

- **Ruehr S**, Citrini A, Wendland E, Dukes J, Rosa L. Double cropping increases water scarcity and irrigation demand in Brazilian soybean croplands.
- Dannenberg M, Barnes M, Biederman J, Johnston M, Meerdink S, **Ruehr S**, Scott R, Smith W, Williams P. Improved estimation of the interannual variability of dryland carbon and water fluxes.

In preparation

- Cabiyo B, **Ruehr S**, Arora T, Nolan CJ, Kueppers L, Field C. The durability of forests in a changing climate. (Target journal: *Nature*).
- **Ruehr S**, Rosa L. Biofuel's potential in United States constrained by future drought. (Target journal: *Nature Energy*).

Publications from consulting

- Rutkove SB et al. [6 authors incl. **Ruehr S**]. 2022. Design and pilot testing of a 26-gauge impedance-electromyography (iEMG) needle in wild type and ALS mice. *Nerve & Muscle*, 65, 6. 10.1002/mus.27551.
- Chin A, **Ruehr S**, Tarulli A, Rutkove SB. 2007. Saline-saturated Balsa Wood as a Testing Medium for Rotational Electrical Impedance Myography. *IFMBE Proceedings*, 17, 272-275. 10.1007/978-3-540-73841-1_72.

PRESENTATIONS

First author oral presentations

- Multiple cropping increases water scarcity and irrigation demand in Brazilian croplands, May 2026, European Geophysical Union, Emerging interdisciplinary perspectives in water resilience and risk science: society–ecology–freshwater nexus approaches and socio-meteorology, Vienna, Austria.
- Decoding ecosystem drought responses: From physiology to resilience across natural and managed landscapes (**invited**), May 2026, Biological Sciences & Scientific Imaging and Computing Institute, University of Utah, UT.
- AI for climate resilience: improving monitoring, understanding and smart management across scales (**invited**), April 2026, School of Earth & Sustainability, Northern Arizona University, AZ.
- Ecohydrology insights for water resource management in agroecosystems (**invited**), December 2025, American Geophysical Union Fall Conference, Frontiers in Ecohydrology.
- Understanding and supporting resilient ecosystems (**invited**), August 2025, Lawrence Livermore National Lab, Livermore, CA.
- When in drought: understanding and supporting resilient ecosystems (**invited**), April 2025, Columbia Climate School, NY.

- Terrestrial drought resilience across scales: advancing monitoring, understanding, and management (**invited**), April 2025, Lamont Observatory, Columbia University, NY.
- Emerging satellite products unveil cropland water use efficiency trends and drivers in California's Central Valley, December 2024, American Geophysical Fall Meeting, Washington DC, USA.
- Evidence and attribution of the land carbon sink's historic enhancement (**invited**), October 2023, Max-Planck Institute for Biogeochemistry, Jena, Germany.
- Groundwater drought decreases carbon fixation in a semi-arid oak savannah, September 2023, CREAM, Barcelona, Spain.
- Hyperspectral imagery illuminates drivers of solar-induced fluorescence across landscapes, December 2022, American Geophysical Fall Meeting, Washington DC, USA.
- Groundwater drought decreases carbon fixation in a semi-arid oak savannah, December 2022, American Geophysical Fall Meeting, Chicago IL, USA.
- Quantifying ecosystem reliance on groundwater, December 2021, American Geophysical Fall Meeting, New Orleans LA, USA.
- Carbon emissions and offsets: Global and local research (**invited**), August 2021, Distinguished Speaker at ARCS Forward National Speaker Series.
- Incorporating stakeholder feedback in paleoclimatology research, 2019, Woods Hole Oceanographic Institution MA, USA.
- Tracing ancient cyclones: paleoclimate, oral history & climate futures (**invited**), November 2018, University of the South Pacific Emalus Campus, Vanuatu.
- The oasis effect: evaluating intrinsic biophysical mechanism theory and its implications for sustainable water management in Zhangye, Gansu, China, May 2018, Yale University CT, USA.
- Intrinsic Biophysical Mechanism Theory & the Oasis Effect, March 2018, Key Laboratory of West China's Environmental System, Lanzhou University, Gansu, China.
- Exploring the Oasis Effect with Land Surface Modeling (**invited**), March 2018, School of Geography, Beijing Normal University, Beijing, China.

First author posters

- Evidence and attribution of the land carbon sink's historic enhancement, December 2023, American Geophysical Union Fall Meeting, Chicago IL, USA.
- Picturing SIF: field readiness and initial results from a novel SIF imaging instrument, December 2021, American Geophysical Fall Meeting, New Orleans LA, USA.
- Latent heat drives cooling over oases, December 2020, American Geophysical Fall Meeting.
- Newfound aspects of ancient hurricanes: reconstructing storm intensity and sediment deposition dynamics in northeastern coastal ponds, August 2017, Woods Hole Oceanographic Institution, MA, USA.

MEDIA & OUTREACH

Science communication

2023 [FLUXNET blog](#)
 2022 [Keenan Group TikTok](#)

2022 [AmeriFlux 25 years data visualization tool](#)
2022 [Berkeley Science Review](#)
2019-2020 [Provincetown Independent](#)
2019 [InsideClimate News](#)
2010 [WOMR Cape Cod's Outermost Radio](#)
2016, 2018 [Provincetown Banner](#)

Press

2023 [Ask MIT Climate](#)
2023 [Phys.org](#)

WORKSHOPS

2025 Identity-Based Risks in Field Work, UC Berkeley, CA
2024 Center for Climate Sciences Summer School, NASA Jet Propulsion Lab, Pasadena, CA
2024 Spring Teaching Conference, UC Berkeley, CA
2024 FieldFutures Harassment Prevention Training, UC Berkeley, CA
2024 DroneCamp, CSU Monterey Bay, CA
2022 Field Safety Workshop, AmeriFlux
2022 FluxCourse, AmeriFlux at Niwot Ridge Long-term Ecological Research Program, CO